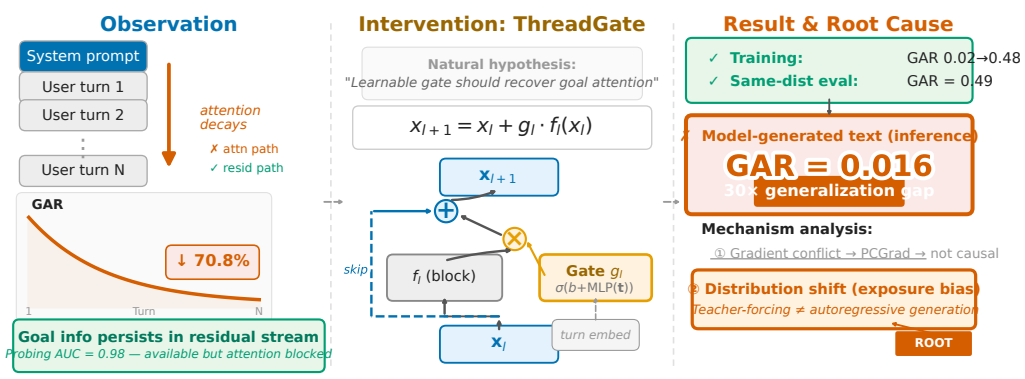
Observation
System prompt
User turn 1
User turn 2
User turn N
attention decays
✗ attn path
✓ resid path
GAR
↓ 70.8%
1
Turn
N
Goal info persists in residual stream
Probing AUC = 0.98 — available but attention blocked
Intervention: ThreadGate
Natural hypothesis:
"Learnable gate should recover goal attention"
$x_{l+1} = x_l + g_l \cdot f_l(x_l)$
x_{l+1}
skip
f_l (block)
Gate g_l
$\sigma(b+\text{MLP}(\mathbf{t}))$
turn embed
x_l
Result & Root Cause
✓ Training: GAR 0.02→0.48
✓ Same-dist eval: GAR = 0.49
✗ Model-generated text (inference)
GAR = 0.016
30× generalization gap
Mechanism analysis:
① Gradient conflict → PCGrad → not causal
② Distribution shift (exposure bias)
Teacher-forcing ≠ autoregressive generation
ROOT